LECTURE 1

REPRESENTATIONS

What is a representation?

PRESS → OR SPACE.

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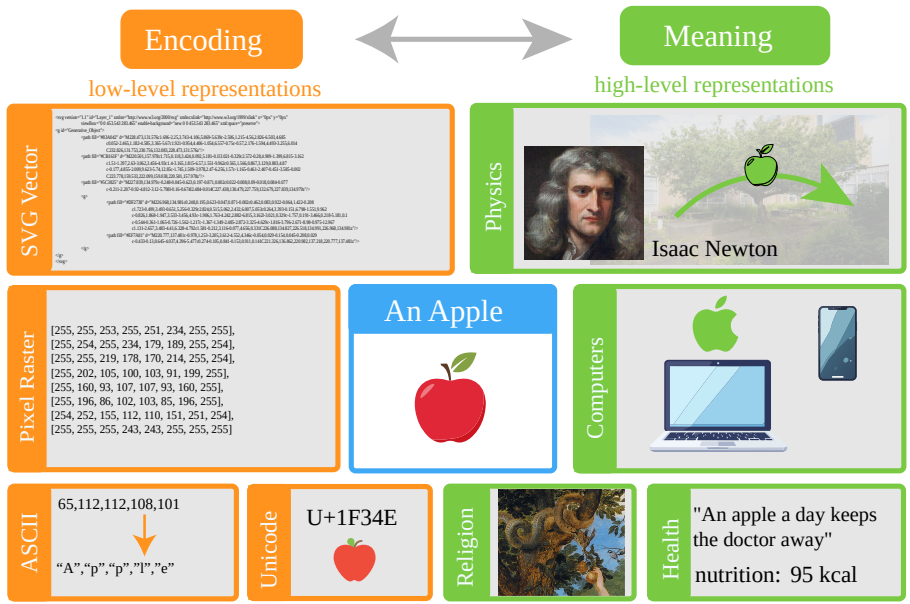
UNIVERSITÄT BONN

What is an Apple?

Apple by Fir0002/Flagstaffotos license CC-BY-NC, Earth added via ChatGPT

Describe an Apple.

Meaning and Encoding of Representations



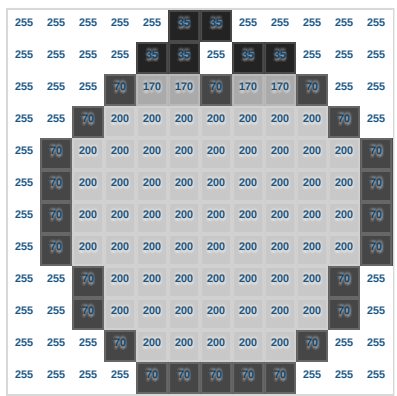
The same object can be encoded in many ways. A good encoding representation makes useful structure visible and computation easier.

But representation is not only encoding.
Representation is also meaning.

Choosing the right Encoding matters

Examples

Images



Math

XLVII + LXXVIII = ?

47 + 78 = 125

Places

COORDINATES

50°43'57.73" N,
7°06'16.63" E

PLACE

Hofgarten, Bonn, Germany

An Encoding does not carry Meaning.

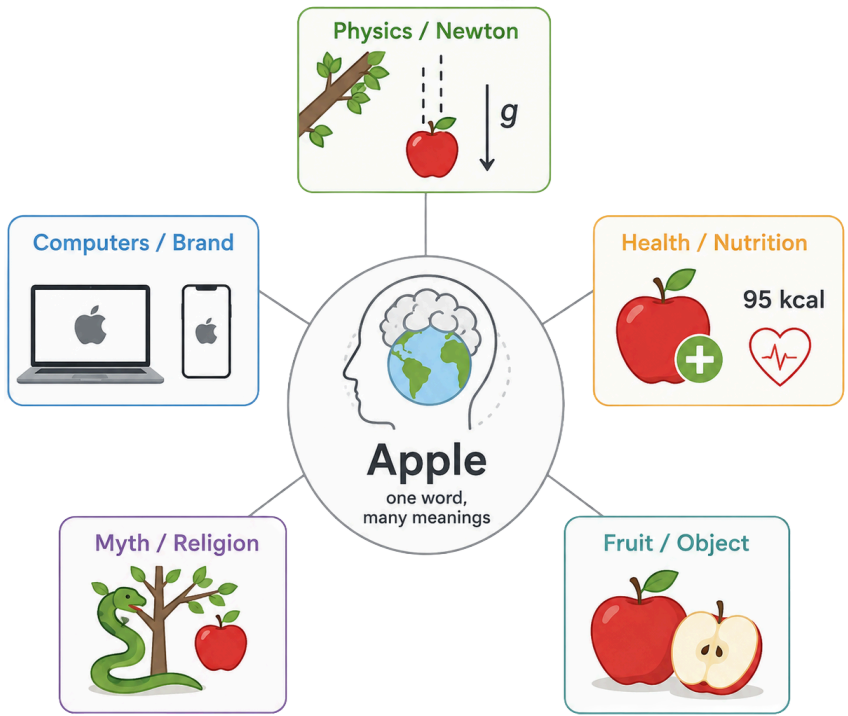


Image created with ChatGPT

This Lecture: Ideas and Methods

Philosophy

What do we mean by a representation?

- How does our mind represent things and places?
- Encoding vs. meaning
- Re-thinking ideas we take for granted

Technology

How do computers represent the world?

- Computer vision
- Natural language processing
- Math: vectors and similarity
- Methods: PCA, t-SNE, embeddings

Learning outcomes

Discuss representations of the same object and **choose** a suitable one for a problem.

Not comfortable — Very comfortable

Click to respond

Discuss the similarities and differences in visual, word, or geospatial representations.

Not comfortable — Very comfortable

Click to respond

Understand how cosine similarity measures how close two embeddings are.

Not comfortable — Very comfortable

Click to respond

Understand how PCA and t-SNE visualize high-dimensional vector representations.

Not comfortable — Very comfortable

Click to respond

Describe how tokenization, embeddings, and next-token prediction represent language.

Not comfortable — Very comfortable

Click to respond

Understand what geospatial information is encoded in geospatial embeddings.

Not comfortable — Very comfortable

Click to respond

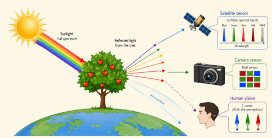
Lecture Outline

What is a representation?

Representations

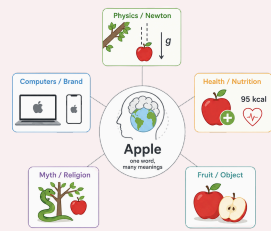
What are representations?

Vision



How vision encodes the world

Language



How text becomes meaning

Spatial

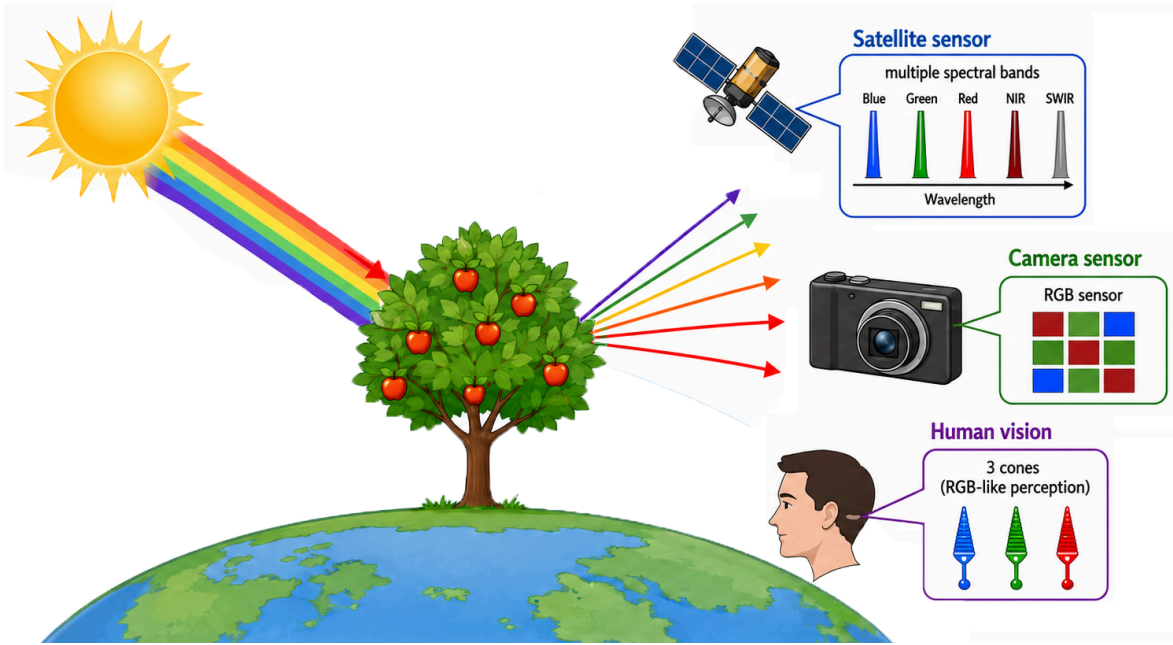
How places carry spatial meaning

Die Stadt in meinem Kopf. (2016). *Süddeutsche Zeitung*.

Visual Representations

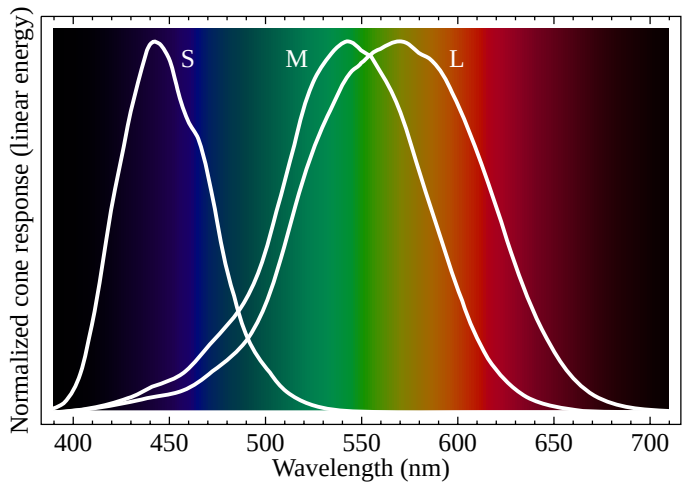
We perceive our world in representations

Example: Vision

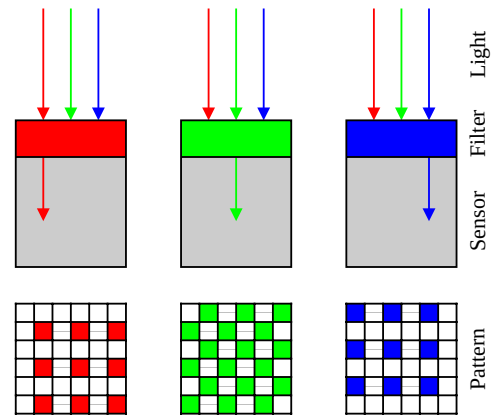


Encoding Light to Meaning

Sensitivity of cone cells (S, M, L) in human eyes

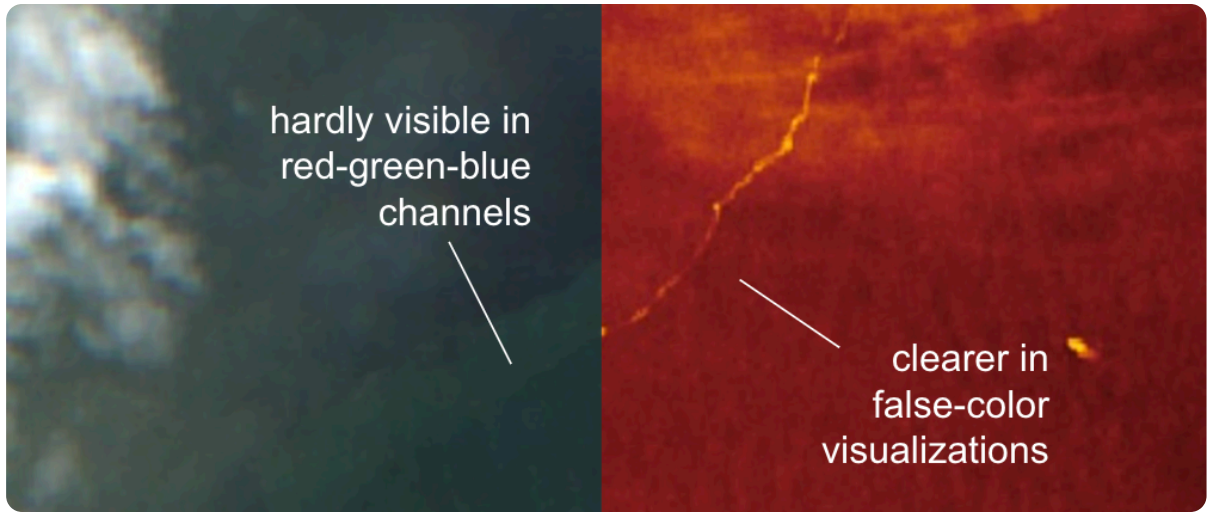


Bayer Filter on Cameras



Cameras are built to mimic our eyes: More green pixels in CCD camera mimics the color sensitivity in our eyes.

Sensors can Measure different wavelengths



Example: plastic litter and marine debris (Sentinel-2)

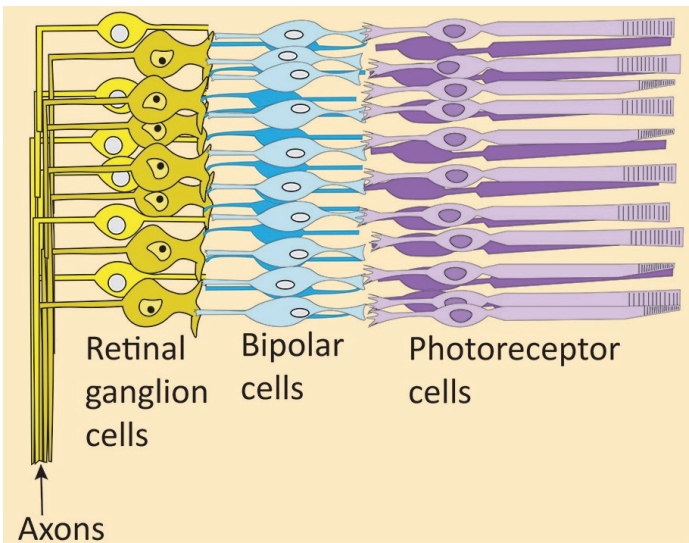
Other Species See the World Differently.

A useful representation depends on the problem we are trying to solve.

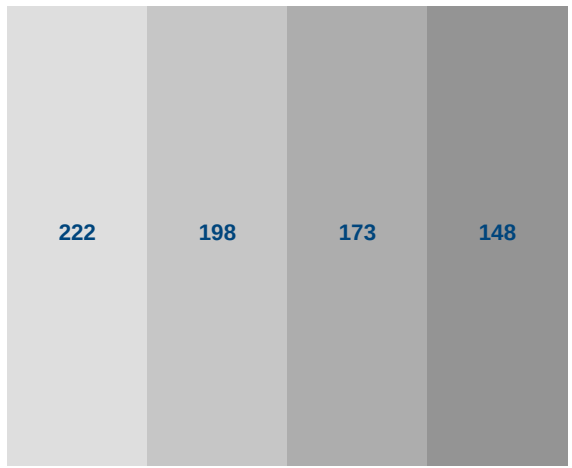
Bees see ultraviolet patterns that humans miss, because their visual system is tuned to different tasks.

Our Vision is Edge-enhanced

Retinal cells implement an edge filter through lateral inhibition.



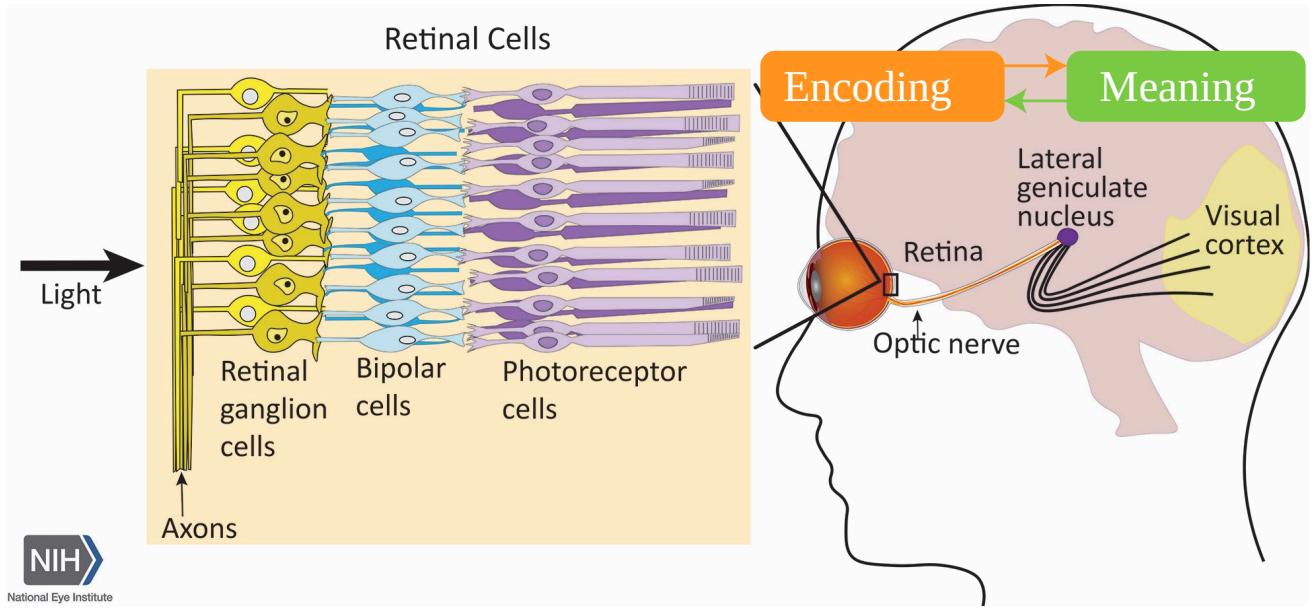
Mach Bands - Optical Illusion (Mach, 1865)



Our eyes are not objective, they capture an edge-enhanced representation of the world.

Shapley, R. M., & Tolhurst, D. J. (1973). Edge detectors in human vision. *Journal of Physiology*, 229, 165-183.

Our mind selects what we perceive



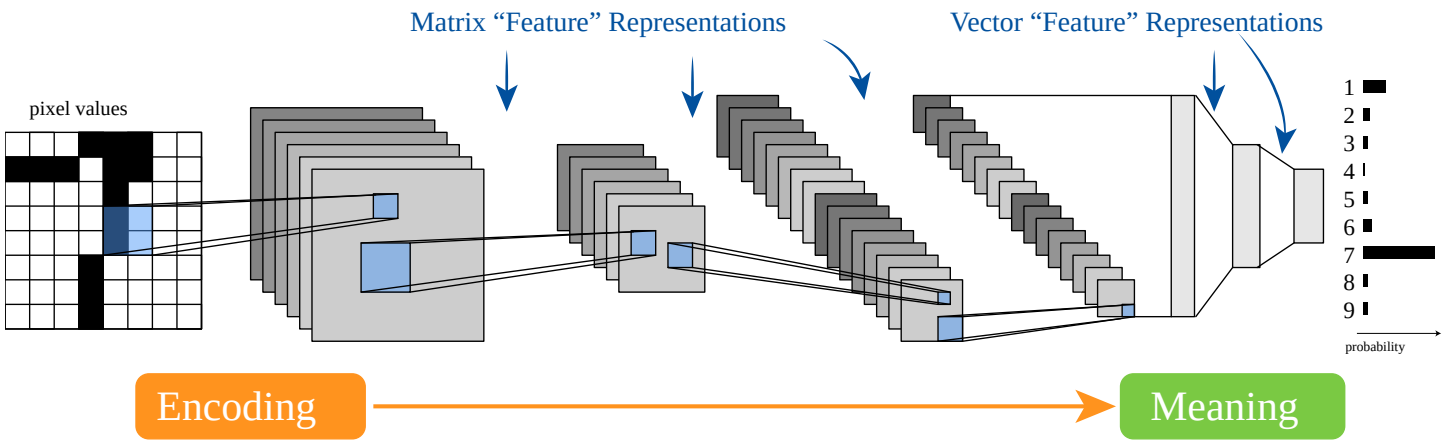
Yang, J., et al. (2025). Differential impact of retinal lesions on visual responses of LGN X and Y cells. *The Journal of Neuroscience*. DOI: 10.1523/JNEUROSCI.0436-25.2025.

Our Vision is Selective

Simons, D. J. (2010). Monkeying around with the gorillas in our midst: Familiarity with an inattentional-blindness task does not improve the detection of unexpected events. *i-Perception*. https://doi.org/10.1068/i0386

Low-Level and High-Level Features in CNNs

From Encoding to Meaning



Excurse: Dimensionality Reduction

To to visualize high-dimensional vector representations

Principal Component Analysis (PCA)

Rotate to PCA projection | Reset | ☑ Show projections

t-distributed Stochastic Neighborhood Embedding (t-SNE)

Run t-SNE projection | Reset | ☑ Show trajectories

Feature Representations in LeNet (2000)

Supervised Learning: From Labels to Loss

Learning from labeled images

- **Input:** a 28×28 image of a handwritten digit
- **Known truth:** the human-provided label `7`
- **Prediction:** probabilities over the ten digit classes
- **Learning:** iteratively adjust the model to minimize the prediction error
- **Goal:** make the probability for the true label `7` as high as possible

`image pixels → CNN representation → class probabilities → label`

> **How do we measure whether the prediction is correct or incorrect?**

Classification example: Cross-Entropy Loss

$$\mathcal{L}_{CE}(y, \hat{y}) = -\sum_{c=1}^{C} y_c \log(\hat{y}_c)$$

For a one-hot target label:

$$\mathcal{L}_{CE} = -\log(\hat{y}_{\text{correct}})$$

Correct class: `7`

- If `p(7) = 0.90` → low loss
- If `p(7) = 0.10` → high loss

> **Cross entropy rewards high probability on the correct class and penalizes confidence in wrong classes.**

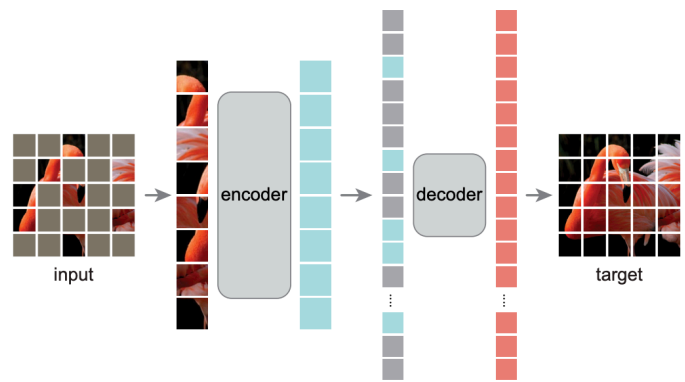
Self-supervised Learning

Contrastive Learning

Identify the corresponding object among two different views of the same object.

Masked Autoencoding

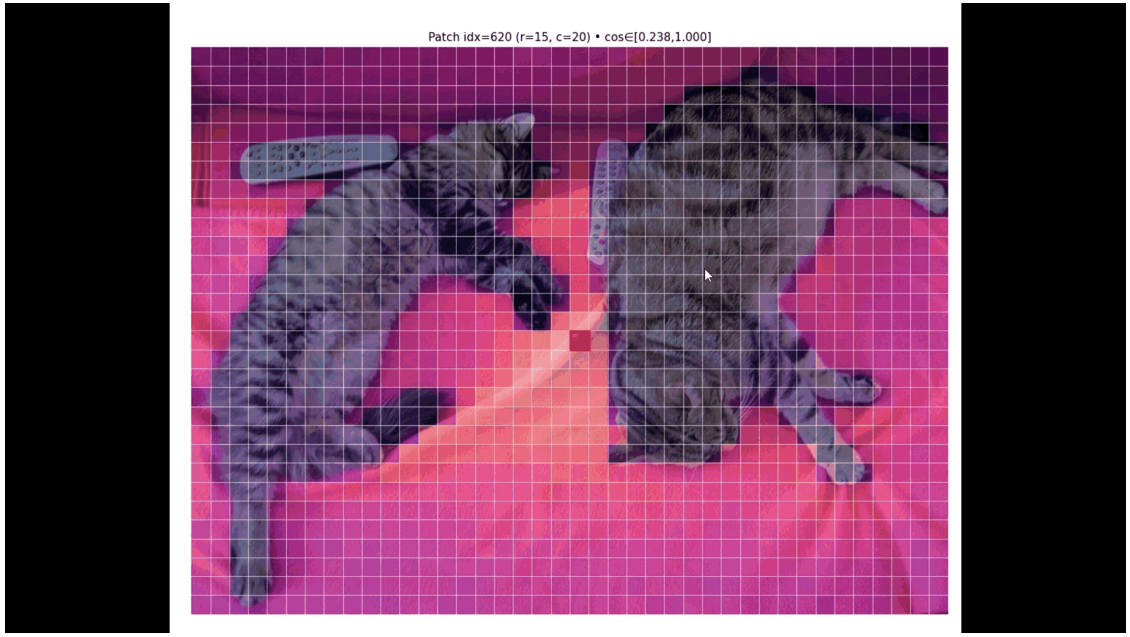
Train the model to correctly inpaint a masked portion of an image.



Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., & Joulin, A. (2020). **Unsupervised Learning of Visual Features by Contrasting Cluster Assignments** . NeurIPS. Code: **facebookresearch/swav**.

He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). **Masked autoencoders are scalable vision learners** . CVPR, 16000-16009.

High-level Visual Features (Dinov3, 2025)



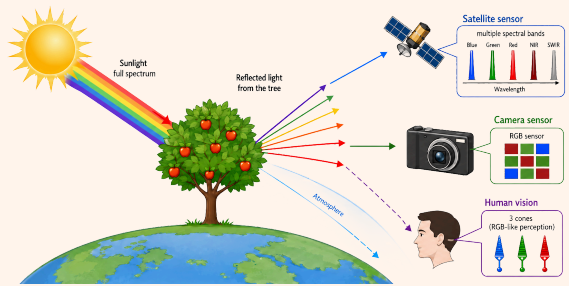
Demo: **devMuniz02/DINOv3-Interactive-Patch-Cosine-Similarity**
DINOv3 model: **Siméoni, O., Vo, H. V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., et al. (2025). DINOv3. *arXiv:2508.10104*.**

High-level Visual Features (SAM 2, 2024)

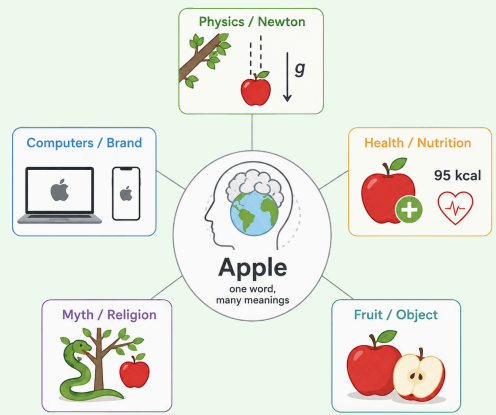
Try the Segment Anything Model (SAM) 2 demo

From Physical to Societal Representations

Observing the Physical World with Vision



Understanding Societies by Language

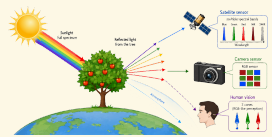


Lecture Outline

Representations

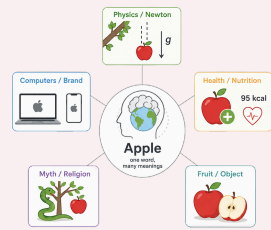
What are representations?

Vision



How vision encodes the world

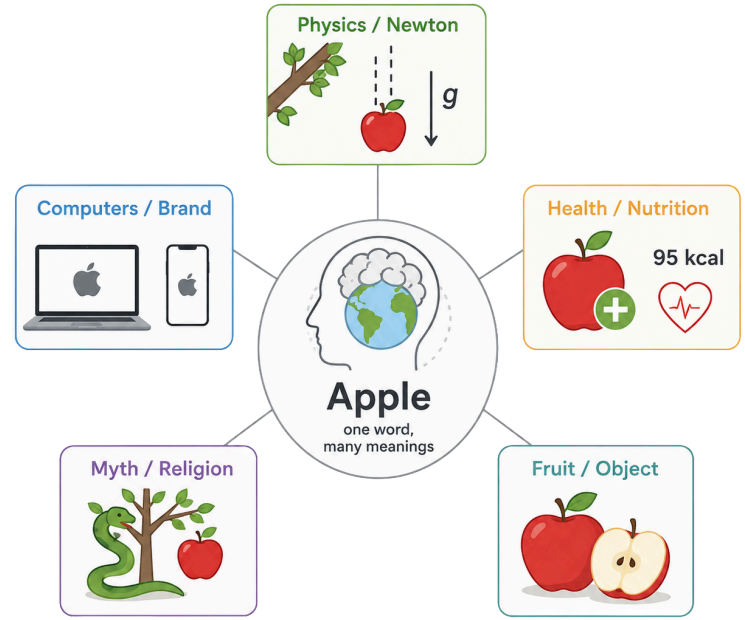
Language



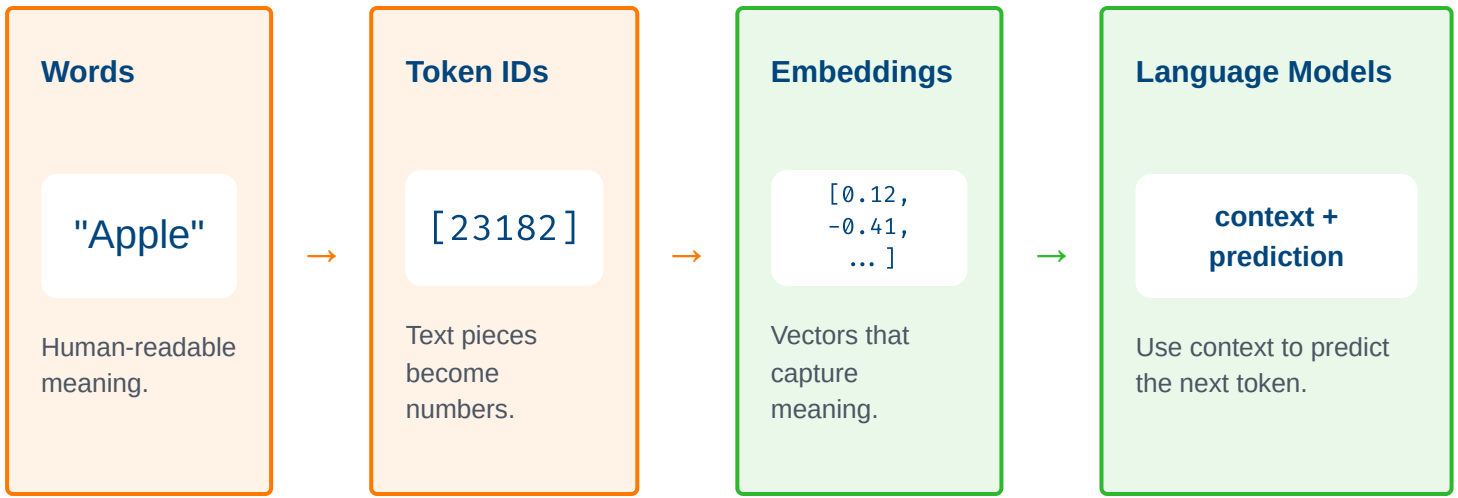
How text becomes meaning

Die Stadt in meinem Kopf. (2016). *Süddeutsche Zeitung*.

Language Representations



From Text Encoding to Language Models



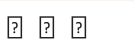
From symbols to meaning in context.

Character Encodings

Human writing systems

Symbols carry meaning.

Cuneiform
c. 3200 BCE



Hieroglyphs
c. 3100 BCE

Chinese characters
c. 1200 BCE - today

Alphabetic writing
c. 700 BCE - today

A B C ...

Digital-era symbols
20th-21st century

@ # :) 😂 🌍

Learned meaning
different fonts, same symbol

A A A A A A A A

Machine character encodings

Computers store characters as numbers.

ASCII
early character numbers

```
A → 65
B → 66
a → 97
```

Unicode code points
global character IDs

```
A → U+0041
ä → U+00E4
🌍 →
U+1F30D
```

UTF-8 bytes
stored byte sequences

```
A → 41
ä → C3 A4
🌍 → F0 9F
8C 8D
```

What does "A" + "B" give in C++? Try cpp.sh

```
std::cout << ('A' + 'B') << std::endl;
```

Word Tokenizer

Language Models tokenize text

Question: How many "r"s are in Strawberry?

Our view	`S t r a w b e r r y`
Model view	`[Str] [aw] [berry]`
Token IDs	`[3504, 1134, 19772]`

Vocabulary: ~100k–200k tokens.

Try it yourself: platform.openai.com/tokenizer

Try: Strawberry · Schifffahrt · Poppelsdorfer Allee

We also read in chunks

Hmuans do not raed ervey wrod letetr by letetr. As lnog as the frist and lsat lettres are rghit, the txet is sitll esay to raed.

We read learned patterns, not single letters.

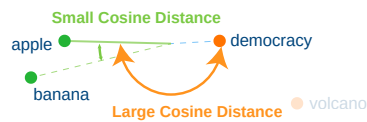
Word Embeddings

From token IDs to vectors

token	ID	vector
strawberry	8123	[0.21, -0.11, ...]
apple	15421	[-0.35, 0.42, ...]
banana	9821	[0.19, 0.05, ...]

IDs are arbitrary. Vectors are learned and capture meaning.

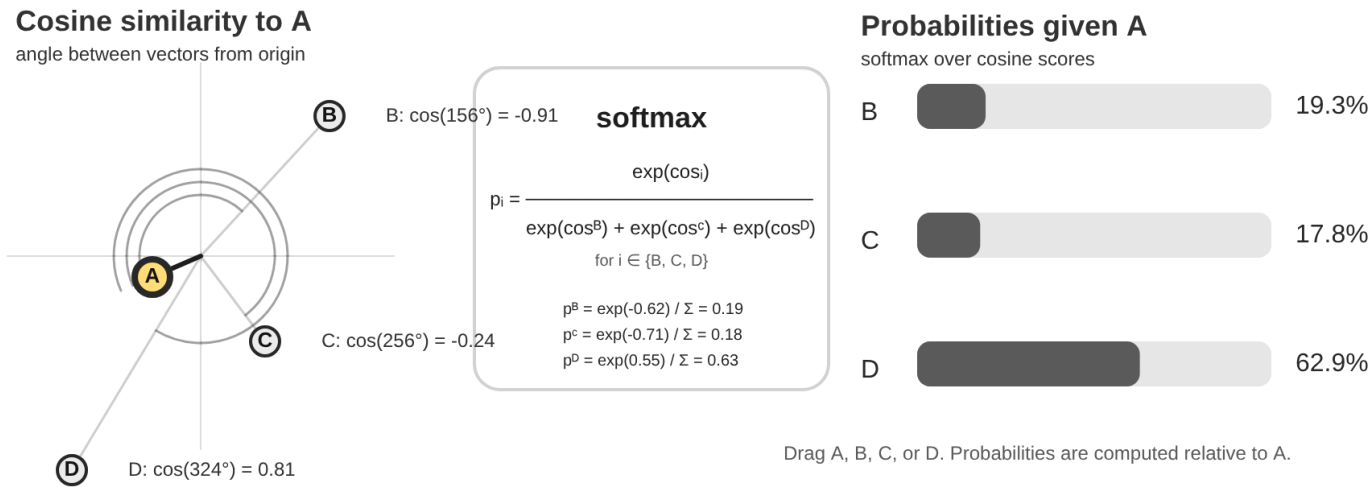
Words are points in a semantic space



Each word is a vector.

Similarity = cosine of the angle.

Softmax: Cosine Similarity to Probabilities



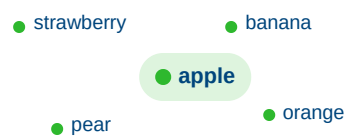
Context matters

Same word, different neighbours.

Fruit context

I ate an **apple**.

2D illustration of high-dimensional embeddings

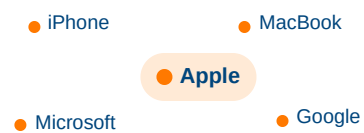


nearby words: fruit, sweet, edible

Technology context

I bought the new **Apple**.

2D illustration of high-dimensional embeddings



nearby words: device, software, company

Language Models predict the next token.

Static embedding

`apple → one vector`

Contextual embedding

`apple → vector depends on context`

Inputs

1. masked word: `<CLS>`
2. context

→

Language model

`p(<CLS> | context)`

→

Output probabilities

```
fruit: 0.42
company: 0.31
device: 0.18
other: 0.09
```

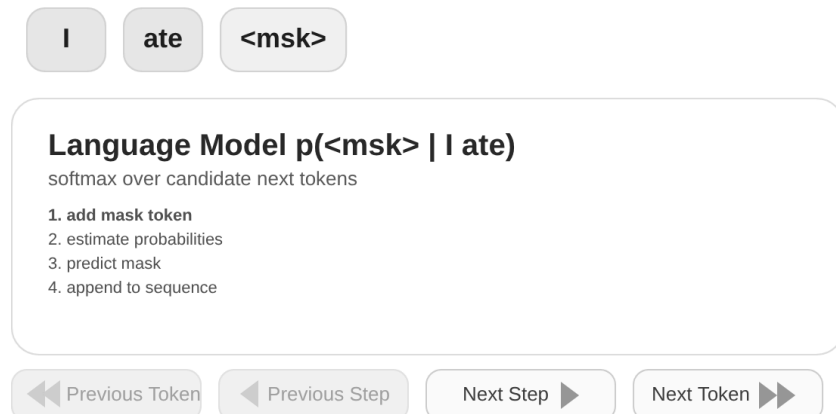
Try it: huggingface.co/spaces/ysdede/fill-mask-demo

Autoregressive next token prediction

One token at a time

Predict the next token, append it, repeat.

> **Autoregressive = one token after another.**



Training signal

Trained to predict the actual next token (cross-entropy loss).

Language Model Training

Pre-training

Predict the next token on huge text corpora.

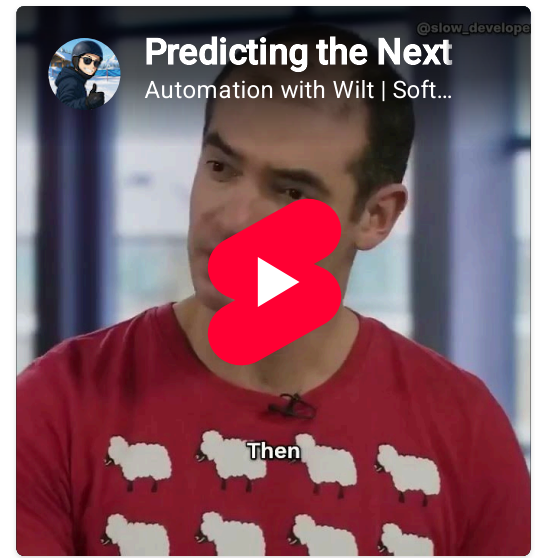
> **Predicting the next token requires world knowledge.**

Fine-tuning

Then tuned to follow instructions.

> **→ AI assistants like ChatGPT, Claude, Mistral.**

More details: **Andrej Karpathy, “Deep Dive into LLMs like ChatGPT”**



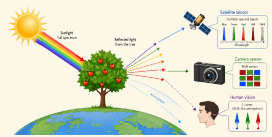
NVIDIA. (2023). *Fireside Chat With Ilya Sutskever and Jensen Huang: AI Today and Vision of the Future.* YouTube short edited by @automationwithwilt .

Lecture Outline

Representations

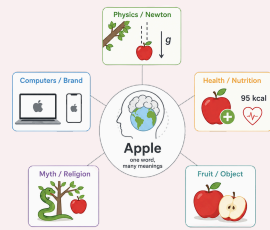
What are representations?

Vision



How vision encodes the world

Language



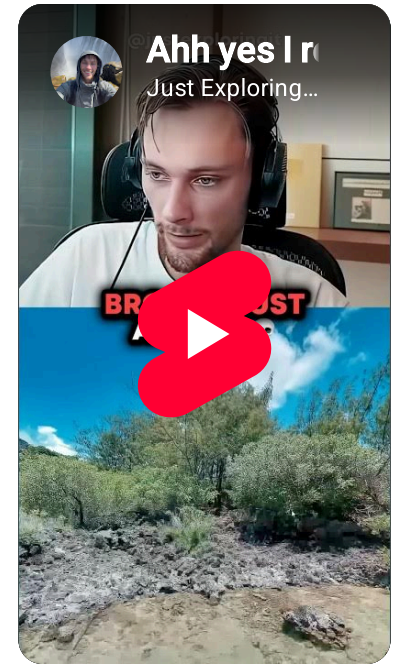
How text becomes meaning

Spatial

How places carry spatial meaning

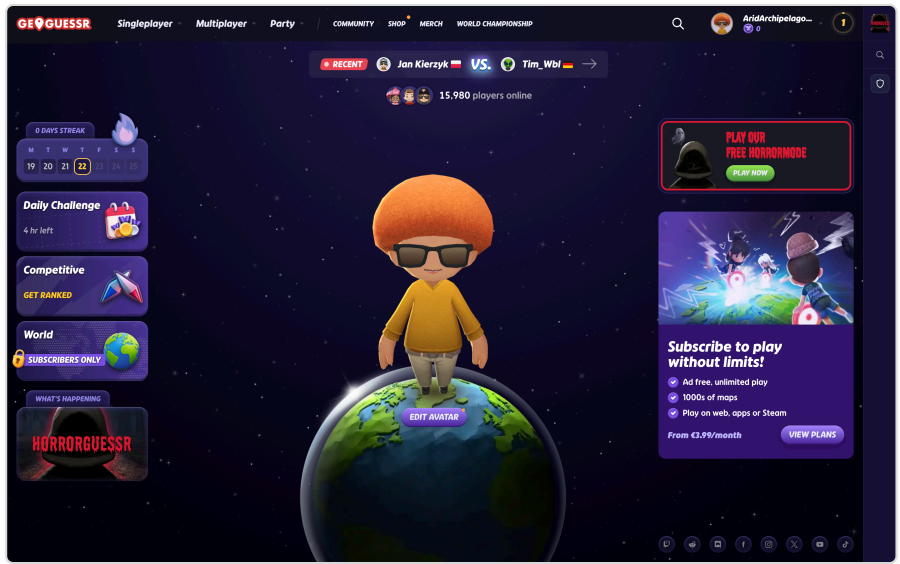
Die Stadt in meinem Kopf. (2016). *Süddeutsche Zeitung*.

Geospatial Representations



Video: Just Exploring It, via YouTube Shorts

GeoGuessR



📍 Play GeoGuessR

Screenshot: geoguessr.com

Geolocalization to Learn a Mental Map

Vision

- regular city blocks,
- gabled slate roofs,
- dense housing
- irregular layout

Geo localization

- Gründerzeit, mid-19th century
- Mid-large sized city
- Remnants of a 18th
- century citadel

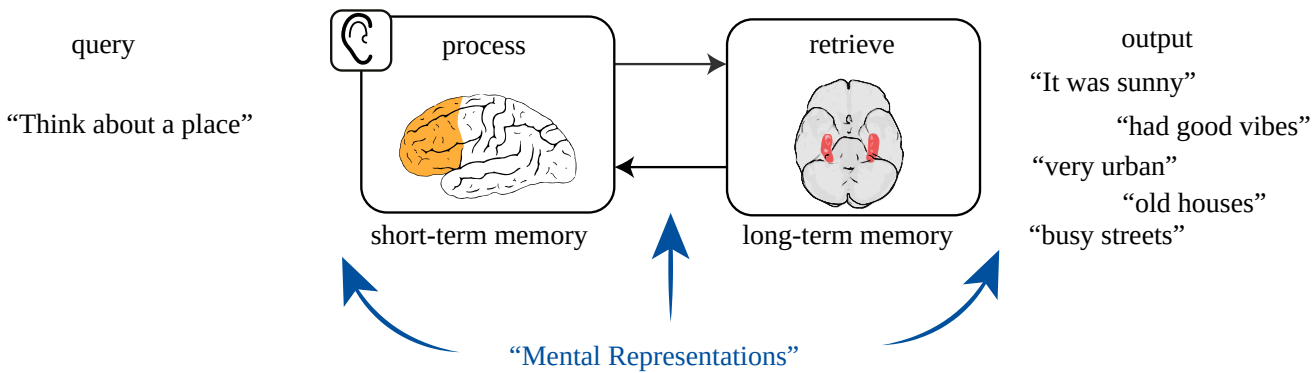
Location

(Mental) Mapping or Context

Mental Representation of a Place

Think of a place you visited on vacation.

Not the country or city name first — think of the actual place. What do you remember?

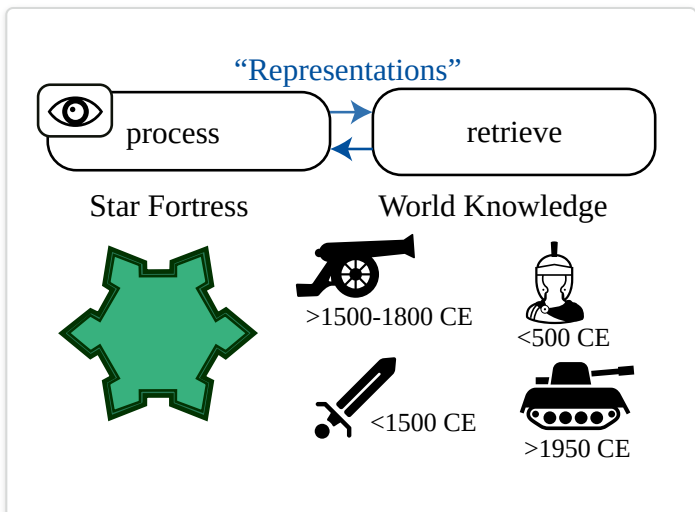


We feel mental representations as "intuitions" and "memories" of things/places.

Preston, A. R., & Eichenbaum, H. (2013). Interplay of hippocampus and prefrontal cortex in memory. *Current Biology*, 23(17), R764-R773.

History: When was this city founded?

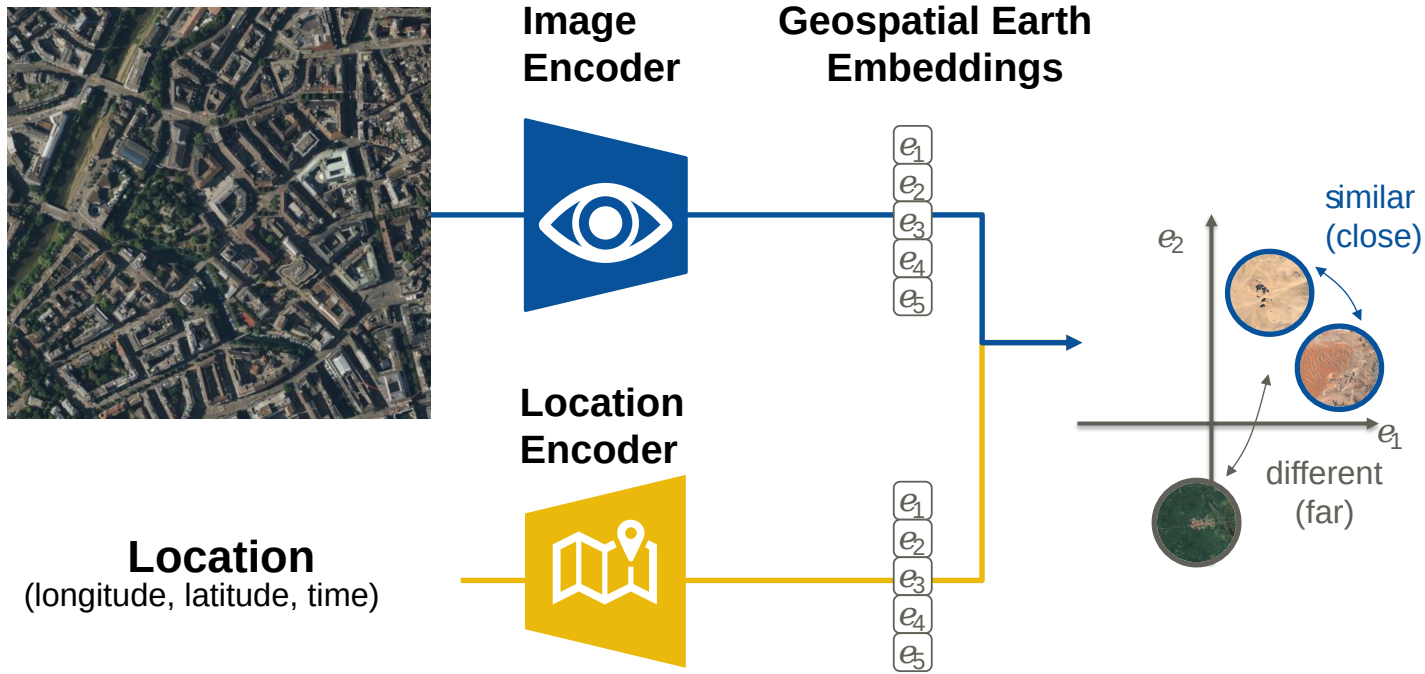
- [ ] 282 CE (Roman/Antique)
- [ ] 1153 CE (Midieval)
- [x] **1593 CE (Renaissance)**
- [ ] 1975 CE (Modern)



Biodiversity: Which Agricultural Landscape supports more Diverse Bird Populations?

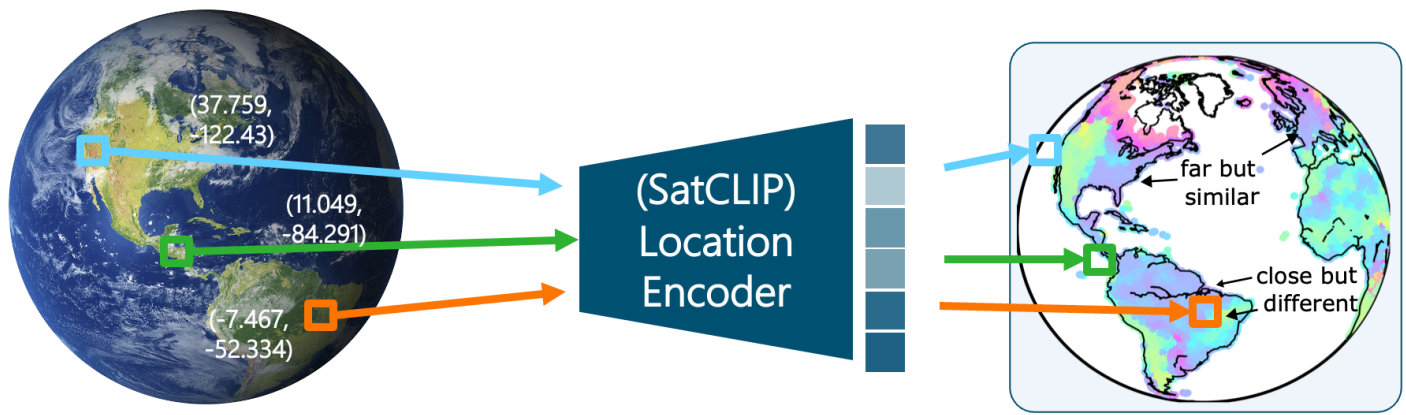
Engist, D., Finger, R., Knaus, P., Guélat, J., & Wuepper, D. (2023). Agricultural systems and biodiversity: evidence from European borders and bird populations. *Ecological Economics*, 209, 107854.

Learning a Map Representation



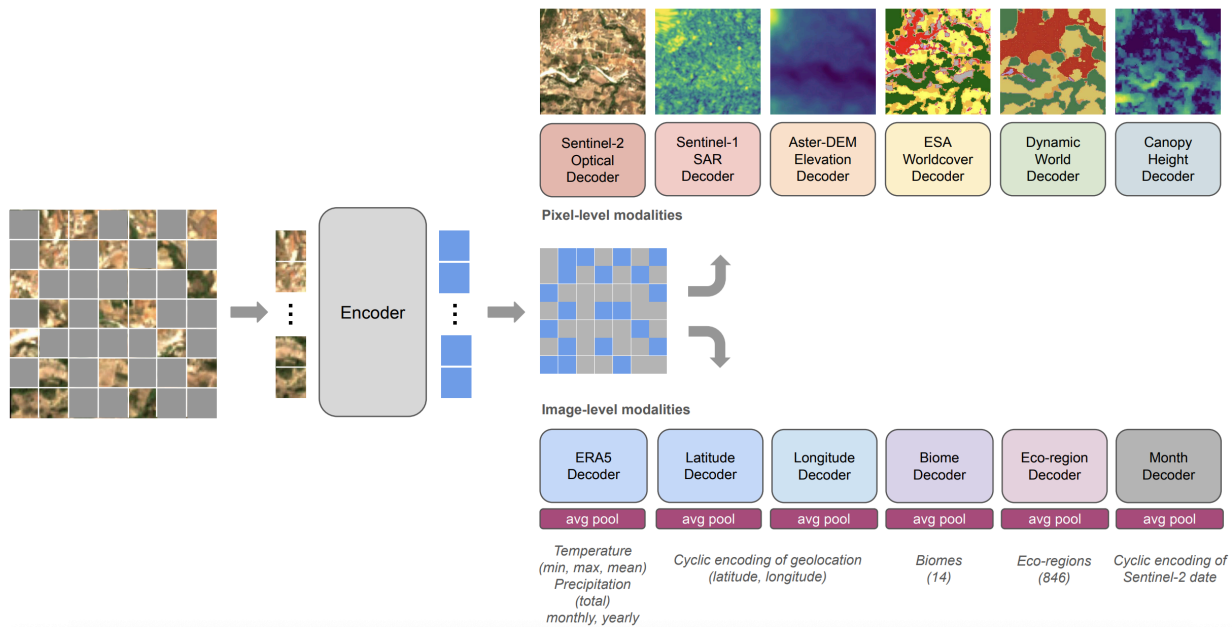
Satellite Contrastive Location-Image Pre-training (SatCLIP)

SatCLIP Encoder for Implicit Embeddings

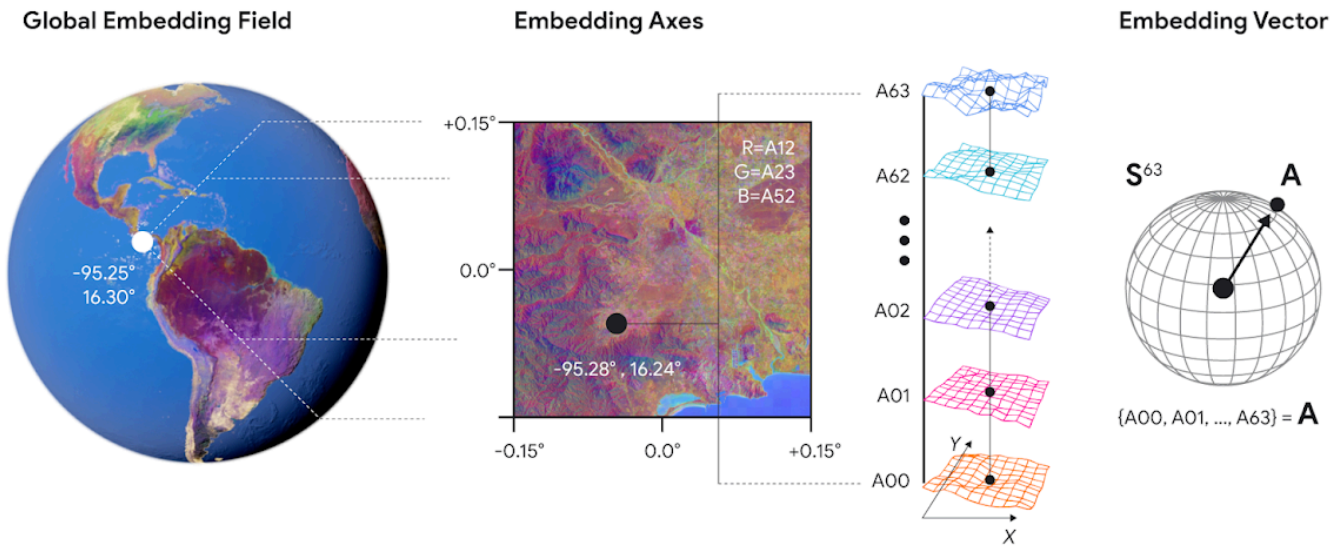


Geospatial Foundation Models

Encode a place through Deep Models

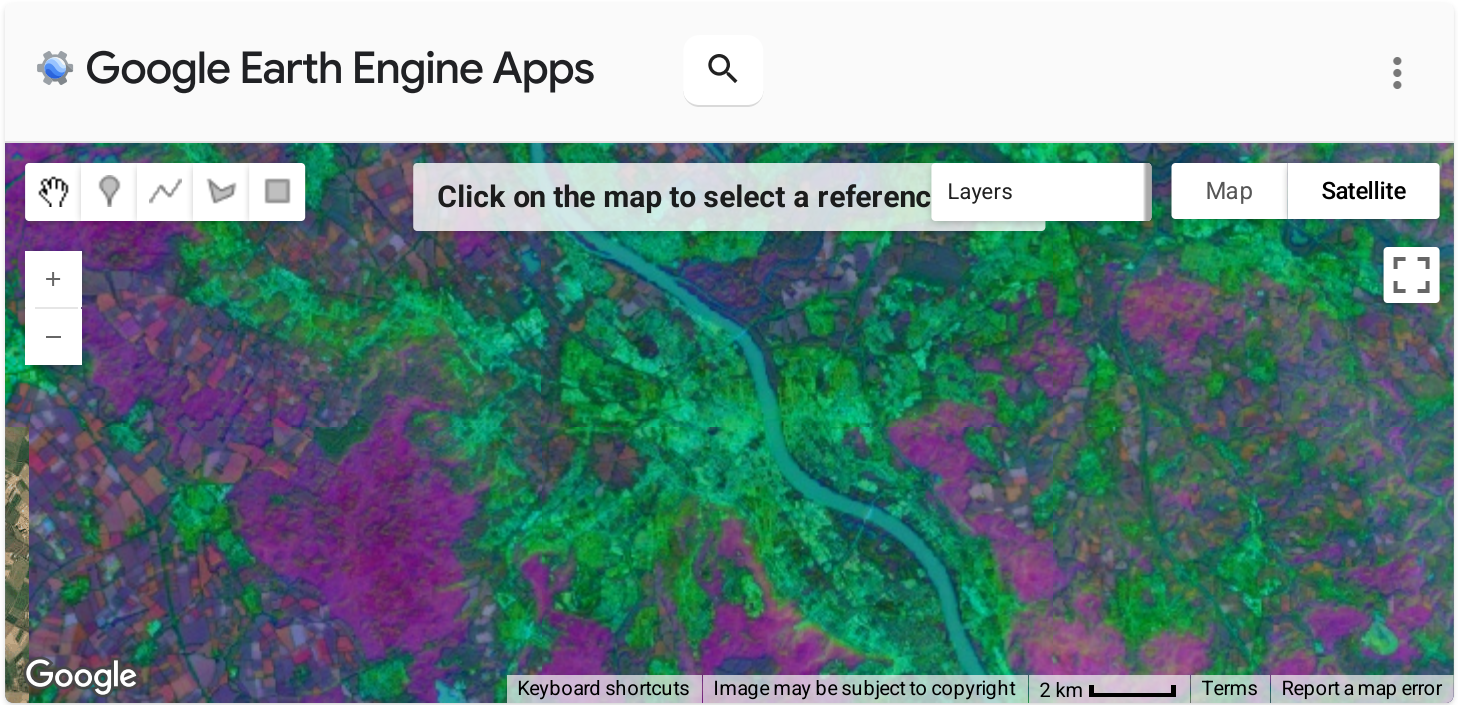


Alpha Earth Embeddings

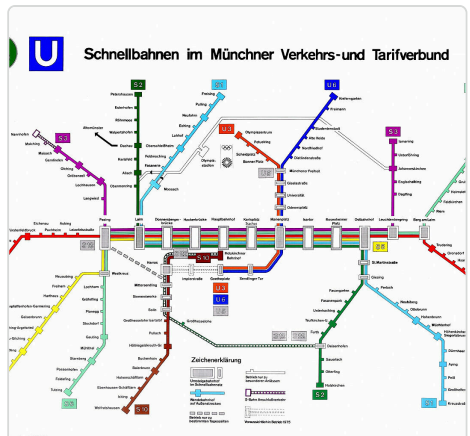


Brown, C. F., Kazmierski, M. R., Pasquarella, V. J., Rucklidge, W. J., Samsikova, M., Zhang, C., ... & Kohli, P. (2025). Alphaearth foundations: An embedding field model for accurate and efficient global mapping from sparse label data. *arXiv preprint arXiv:2507.22291*.

Alpha Earth Embedding Explorer

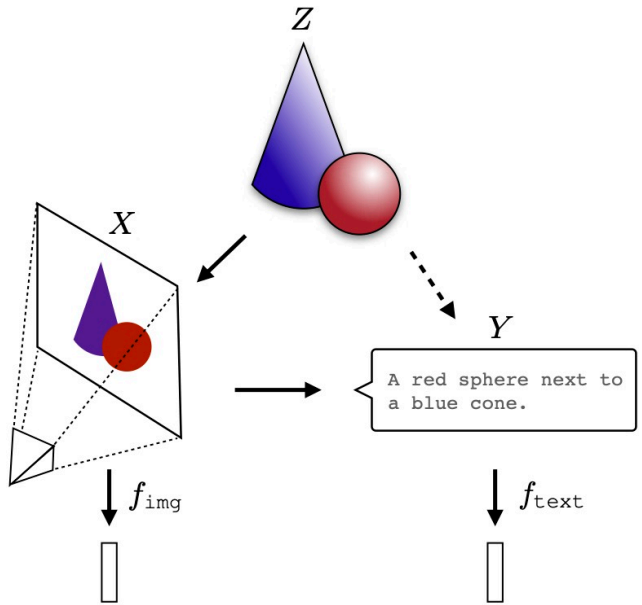


Takeaways - Same city, different maps



We need to select or learn a suitable representation for the problem-at-hand.

The Platonic Representation Hypothesis



Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces.

Huh, M., Cheung, B., Wang, T., & Isola, P. (2024). *The Platonic Representation Hypothesis*. Position paper, ICML 2024.

Revisit the learning outcomes

Light gray fill: your group's before responses today · Click to respond again.

Discuss representations of the same object and **choose** a suitable one for a problem.

Not comfortable — Very comfortable

Click to respond

Discuss the similarities and differences in visual, word, or geospatial representations.

Not comfortable — Very comfortable

Click to respond

Understand how cosine similarity measures how close two embeddings are.

Not comfortable — Very comfortable

Click to respond

Understand how PCA and t-SNE visualize high-dimensional vector representations.

Not comfortable — Very comfortable

Click to respond

Describe how tokenization, embeddings, and next-token prediction represent language.

Not comfortable — Very comfortable

Click to respond

Understand what geospatial information is encoded in geospatial embeddings.

Not comfortable — Very comfortable

Click to respond

See you at 13:00 for Lab 1!

Visualizing high-dimensional representation spaces

We will investigate how **PCA** and **t-SNE** reveal the structure of embeddings.

13:00–15:00 · Seminarraum 2.003
Nußallee 1